

CAELAN HARTLEY

Software Engineer · Backend, Full-Stack, Cloud & AI/ML

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SUMMARY

Software Engineering student (graduating June 2027) with three co-op terms: enterprise banking modernization at CIBC, agentic AI development and large-scale technical project coordination at Kyndryl, and data operations at cybersecurity company BlackBerry. Builds full-stack systems end to end: Python, Java, and Node.js backends, React/TypeScript frontends, multi-agent LLM systems, RAG, PyTorch deep learning models, and containerized deployments on Azure, OpenShift, and Linux.

EDUCATION

Bachelor of Engineering Science (BESc), Software Engineering
Western University · GPA 3.7/4.0 · Dean's Honour List

Expected June 2027
London, ON

TECHNICAL SKILLS

Languages	Python, Java, TypeScript, JavaScript, SQL, Bash
Backend & Web	FastAPI, Flask, Spring Batch, Node.js, Express, React, Pydantic, SQLAlchemy, REST APIs, OAuth 2.0, JWT
Cloud & DevOps	Microsoft Azure (App Service, Blob Storage, Container Registry), Red Hat OpenShift, Kubernetes, Docker, Linux, GitHub Actions, CI/CD, Nginx
Generative AI	Multi-agent orchestration, tool calling, Retrieval-Augmented Generation (RAG), Azure AI Search, Model Context Protocol (MCP), Prompt Engineering, LangChain, human-in-the-loop guardrails, speech-to-text/TTS
Machine Learning	PyTorch, scikit-learn, pandas, NumPy, LightGBM, Hugging Face, transfer learning, feature engineering, CUDA GPU training, model evaluation (ROC-AUC, F1)
Data	PostgreSQL, MongoDB, Azure Data Lake Storage, PySpark, Elasticsearch
Monitoring & Testing	Prometheus, Sentry, pytest, Playwright, Agile/Scrum

EXPERIENCE

Application Developer Co-op CIBC · Private Wealth Management Technology	Sep 2026 – Dec 2026 Toronto, ON
- Contributing to a team-wide initiative modernizing legacy wealth-management systems onto Red Hat OpenShift on Azure, developing batch jobs with Spring Batch (Java) scheduled as OpenShift CronJobs; onboarding to Azure Data Lake and PySpark - Supporting the like-for-like migration of legacy shell-script batch jobs and online transactions (authentication, account data, transfers), helping preserve compatibility with downstream consumers - Working within the team's layered microservice architecture of business, data-access, and backend integration services built on XML/XSD message contracts - Assisting with migration validation through legacy-vs-migrated output comparison and SIT/UAT support, following the team's enterprise CI/CD and SDLC processes (Artifactory, HashiCorp Vault)	
Application Modernization & Cloud Intern Kyndryl · Cloud, Data & AI	May 2025 – Aug 2026 Toronto, ON
- Independently built and deployed Caeprabank, an AI-native banking and insurance platform (React, FastAPI, PostgreSQL), owning Linux-based development, Docker containerization, and deployment on Azure App Service and Blob Storage - Built voice-first claim submission from mobile or desktop using OpenAI Whisper and TTS, with GPT-4o vision assessing damage in claim photos and RAG over policy documents (Azure AI Search embeddings) answering customer policy questions - Built an adjuster portal where insurers review and edit claims, record audio, and get AI transcripts and suggestions - Automated loan and mortgage approval and denial by pairing deterministic business rules with Azure OpenAI reasoning; demoed the platform, led by the insurance module, to internal partners and external prospects - Coordinated a VM-to-cloud migration of 1,000+ endpoints across AIX, Linux, and Windows: joined client calls, led documentation, and contributed to architecture research and design - Built a pricing calculator (Python, React, PostgreSQL) for the managed patching service, used by consulting partners to price real engagements	

- Built an internal SOW Builder that drafts new statements of work using RAG over past SOWs in Azure AI Search

Site Reliability Engineering Intern

May 2024 – Aug 2024

BlackBerry · Data Operations

Waterloo, ON

- Monitored databases, **Elasticsearch** clusters, and infrastructure with **Prometheus**, building alerts and metric exporters
- Automated backups and data-validation workflows with cron jobs and SQL scripts, improving operational reliability

Software Developer

Oct 2022 – Apr 2025

Western Engineering Concrete Canoe Club (W.E.C.C.A.)

London, ON

- Contributed to finishing and optimizing the club's website (HTML, CSS, JavaScript) and PADDL, a JavaFX safety-analysis tool

PROJECTS

my-crm | AI-Native Personal Operations Platform

Solo project

Tech: Python, FastAPI, PostgreSQL, SQLAlchemy, React, TypeScript, Claude API, Plaid, Gmail and Google Calendar APIs

- Built an agentic platform where a custom Claude tool-use orchestrator routes plain-English requests to specialized agents for finances, email, calendar, fitness, and stock options, backed by a **319-test** pytest suite
- Integrated Plaid across multiple bank and credit accounts with incremental sync and a rule-based categorization engine that nets out transfers between my own accounts and credit card payments, so spending is never double-counted
- Designed human-in-the-loop approval gates: the agent must propose and wait for sign-off before sending emails or creating calendar events, with every tool call logged to a live audit trace and sensitive data encrypted at rest
- Integrated **OptionsEdge AI**, my own agent that blends deterministic rules with an LLM to pick two options contracts daily, paper-trades them, and exits via six prioritized rules (stop loss, profit target, trailing stop, time decay, delta, earnings)

Hush | AI-Personalized Romance Audiobook Startup

Independent startup · Pre-launch

Tech: Python, FastAPI, SQLModel, OpenRouter, RunPod Serverless GPU, Docker, GitHub Actions, Sentry, Prometheus, Meta Ads

- Founded and run an independent, production-ready startup (pre-launch) whose web app writes an original story from each reader's preferences and narrates it as an audiobook in **under a minute**
- Orchestrated multiple LLMs through **OpenRouter** in a multi-pass pipeline feeding self-hosted TTS models on **serverless GPU** (parallel chunked rendering, per-chunk retry); shipped via GitHub Actions CD with **457 tests** and security scanning in CI
- Built a per-story cost model (~**140x** variance across voices) to set profitable credit-based subscription tiers; set up launch-ready billing and a Meta Ads pipeline using Claude for ad scripts and Higgsfield for AI video

AI-Generated Face Detector | Deep Learning Image Forensics

Data Science course · Team of 4

Tech: Python, PyTorch, scikit-learn, EfficientNet, Hugging Face, NumPy, pandas, Flask, CUDA, SLURM

- Built a classifier that flags AI-generated faces for identity verification, fine-tuning **EfficientNet-B0** in PyTorch to **~92.5% accuracy** on 20.9K held-out images (validation AUC 0.997)
- Wrote the end-to-end PyTorch training pipeline (augmentation, mixed precision, AUC checkpointing) and scikit-learn baselines (logistic regression, SVM, random forest, KNN), showing a **~27-point** gain from deep features
- Built a Flask dashboard with a REST API to launch and monitor training locally or on an **H100 GPU cluster** via SLURM

Golf Event Planner | AI Tournament Planning Platform

Team of 3

Tech: React, FastAPI, MongoDB Atlas, LangChain, MCP, Docker, Nginx

- Built a platform where organizers plan a tournament by chatting with an AI agent that checks plans against deterministic business rules (RAG, MCP weather tools), then auto-generates tee sheets, rule sheets, F&B orders, and PDF cart placards
- Owned document generation, templated SMTP email, JWT/bcrypt auth, and the MongoDB data layer; deployed with Docker

Assessly | AI Mock Interview Platform

Capstone · Team of 4

Tech: React, TypeScript, Node.js, Express, PostgreSQL, WebRTC, Azure OpenAI, Azure Speech, Azure App Service, Docker

- Built a mock-interview platform in a 4-person Scrum team; integrated the **Azure Text-to-Speech Avatar** over WebRTC with the Azure OpenAI Realtime voice pipeline so candidates talk with an AI interviewer that tailors questions to their resume
- Built email verification, password reset, Blob Storage resume upload, and AI feedback reports; deployed to **Azure App Service**

CERTIFICATIONS & ADDITIONAL

Certifications

Azure Fundamentals (AZ-900) · Azure AI Fundamentals (AI-900) · Anthropic Claude Code in Action

Training

Red Hat System Administration I & II · Kubernetes Application Developer · Spring Batch · Ansible

Hackathon

DataQuest: clinical-trial patient matching using a LightGBM LambdaRank ranker and feature engineering

Languages

English, French (French Immersion)

Interests

Former competitive soccer and hockey player · Non-fiction and self-help books · Fitness · Chess